quotations increases significantly when the tick size is reduced.[1281] Thus the decline in liquidity that we document is only a decline in displayed liquidity. Second, quotes tend to congregate at the price just worse than the quoter's desired price so that the quoter does not lose money on a transaction. When a wider tick is tightened, quotes that were previously congregated at the wide tick will spread out at prices better than the previous tick allowed. Thus, a market participant taking liquidity from multiple price layers in the order book to fulfill an order will have some shares that transact at superior prices than it would have with the wider tick.[1282]

Table 8 also presents the effect of the TSP conclusion on the round-trip cost to transact a trade for 10 round lots (1,000 shares).[1283] This analysis suggests mixed results for the effect of the tick size reduction on the cost of executing a 10-round lot trade. For bin 1 stocks, the total round-trip cost of a 10-round lot trade decreased when the tick size was lowered – suggesting an improvement in liquidity deeper in the book. For stocks in bin 2 (i.e., less tick-constrained stocks), the effect was not clear. The OLS regressions suggested no effect, while the quantile

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1281 See analysis presented in Nasdaq Intelligent Tick Proposal, supra note 150; see also Justin Cox, et al., Increasing the Tick: Examining the Impact of the Tick Size Change on Maker-Taker and Taker-Maker Market Models, 54 FIN. REV. 417 (2019); Amy K. Edwards, et al., The Effect of Hidden Liquidity: Evidence from an Exogenous Shock (working paper Mar. 1, 2021), available at https://ssrn.com/abstract=3766512 (2021) (retrieved from SSRN Elsevier database).

1282 Consider a numeric example. A market with a $0.05 tick is quoting asks of 500 shares at $10.05 and 500 shares at $10.10. An investor wishing to purchase 700 shares would purchase 500 at $10.05 and 200 at $10.10 for a total price of $7,045. If the tick shrinks to $0.01 and cumulative shares posted decline by 20% - for example – but those shares are spread evenly over the finer grid then there would be 80 shares at each price level from $10.01 to $10.10. An investor wishing to buy 700 shares would need to purchase 80 shares at each price level from $10.01 to $10.08 and 60 shares at $10.09 for a total purchase price of $7,034. So even though total depth declined, the cost to execute a 500-share trade would decrease due to more efficiently spreading liquidity across more price levels.

1283 A round-trip trade refers to executing an order to buy or sell the stock and immediately reversing the position with an equal countervailing order. We compute the cost of a roundtrip trade following the methodology laid out in Griffith and Roseman (2019), supra note 1002 and Chung, et al., supra note 1209. The methodology uses MIDAS data to take snapshots of the order book at 15-minute increments throughout the trading day and calculates the transaction costs associated with walking the book up 5 or 25 round lots to execute a large trade.

regressions suggested an increase in trading cost. For stocks in bins 3 and 4 (i.e., those that were not tick-constrained by the $0.05 tick), the effect of lowering the tick size was to increase transaction costs for larger trades. These results cohere with the idea that when stocks are tick-constrained the pricing efficiency made possible by a smaller tick improves liquidity, and for stocks with wider quoted spreads a smaller tick harms liquidity by making individuals less willing to post displayed liquidity due to complexity and the risk of pennying.

In conclusion, the analysis provided here suggests that, for stocks that were limited to just 1-2 ticks intra-spread by the $0.05 tick, the reduction to a $0.01 tick provided an improved trading environment. Thus, trading in an approximate 1-10 tick range intra-spread provided a superior environment to trading in a 1-2 ticks intra-spread range. One caveat here is that the analysis highlights a key tradeoff with a smaller tick for stocks with narrow quoted spreads. They tend to have less depth at the NBBO, but narrower quoted spreads. Thus, the total effect of this tradeoff on execution costs is largely a function of the size of the trade being implemented with smaller trades receiving improved terms while sufficiently large trades get worse terms with a narrower quoted spread. However, as discussed in greater detail in Barardehi et al. (2022), for tick-constrained stocks the point in terms of trade size at which a tick size reduction harms execution quality is quite large, around 50 round lots.[1284] Additionally, for stocks with quoted spreads greater than $0.15, where a $0.01 tick implied more than 15 ticks intra-spread, a $0.05 tick where there were only 3 ticks intra-spread, appeared to provide a superior trading environment. For stocks with quoted spread between $0.10 and $0.15, it is not clear which tick size provided a superior trading environment.

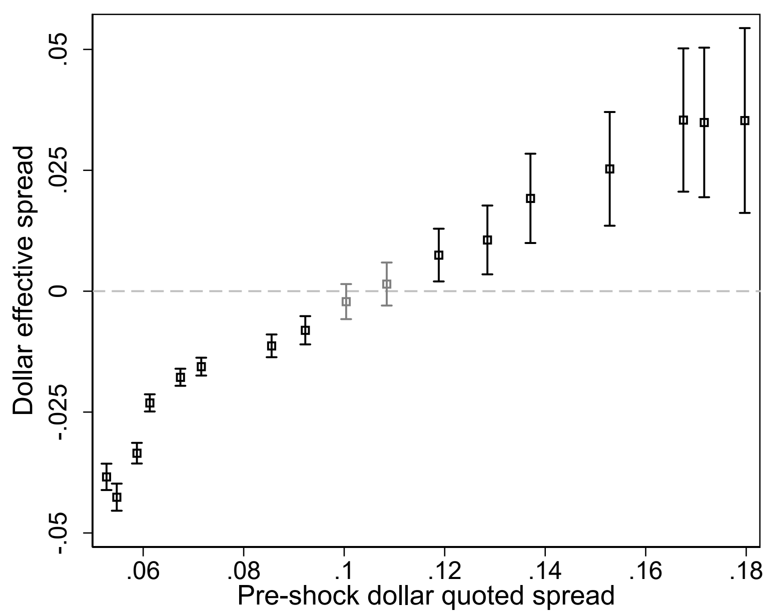
1284 See Barardehi et al. (2022) Table 4. See also Citadel Letter I at 11 requesting an analysis of the joint impact on depth and quoted spread. See also Virtu Letter II at 18 stating that lower quoted spreads would harm markets by reducing the incentive to post liquidity.

In Figure 2, data analysis shows how the quoted spread of a stock during the TSP ("pre-shock dollar quoted spread") correlates with how investor transaction costs, as captured by effective spreads, changed when the TSP ended. For stocks with an average of fewer than two ticks intra-spread (i.e., those with pre-shock quoted spreads of $0.10 or less), a reduction in tick size from 5 cents to 1 cent significantly reduces effective spreads.[1285] For stocks with an average of more than three ticks intra-spread (i.e., those stocks with pre-shock quoted spreads greater

1285 See Proposing Release, supra note 11, at 80322-80323, including Figure 2, which is also in Barardehi et al., supra note 231.

than $0.15), a narrower tick size increases effective spreads. These results are broadly consistent with the findings reported in Table 8.

Figure 2: Difference-in-Difference Effect of the end of the TSP on Dollar Effective Spreads



Source: Stocks in the TSP imposition period 08/12/2016-12/14/2016, where tick size increased from 1¢ to 5¢ for pilot stocks, are grouped into overlapping 6¢ intervals of average May and June 2016, i.e., pre-shock, quoted spreads: {(0,6¢),(1¢, 7¢), . . . ,(15¢, 21¢),(16¢, 22¢). Stocks in the TSP conclusion period 08/08/2018-11/20/2018, where tick size decreased from 5¢ to 1¢ for pilot stocks, are grouped into overlapping 6¢ intervals of average May and June 2018, i.e., pre-shock, quoted spreads. For each intervals, the effect of a tick size change on dollar effective spreads is estimated via difference-in-difference quantile (median) regressions that control for date fixed effects and double-cluster standard errors by stock and date. Point estimates of the treatment effects along with the corresponding 95% confidence intervals are plotted against the median pre-shock quoted spread in the respective interval. See Proposing Release, supra note 11, at 80322-80323, Figure 1. See also Barardehi et al., supra note 221

The Commission's results in Table 8 provide useful information for predicting how the tick size reduction associated with the amendments may affect market quality for stocks priced at, or greater than, $1.00 per share and that receive the $0.005 tick size compared to the current baseline. For stocks with prevailing quoted spreads less than $0.015 there would generally be at most 1.5 ticks intra-spread with a $0.01 tick, or 3 ticks intra-spread with a $0.005 tick. The analysis in Table 8 for bin one stocks suggests that 1-5 ticks intra-spread provides a better trading environment than does just one tick intra-spread.[1286] Additionally, the results for bin 2 stocks suggest that moving from 1-2 ticks intra-spread to 5-10 ticks also generally improves market quality across most measures. In short, the TSP analysis suggests that stocks with fewer than 2 ticks intra-spread on average benefited from a tick size reduction. Consequently, this analysis provides support for the belief that reducing the tick size for stocks that generally have at most 1.5 ticks intra-spread is likely to improve market quality for these stocks. One concern discussed earlier in this section is that a reduction that is too aggressive could harm market quality by providing too many ticks intra-spread. However, the analysis provided in Table 8 does not support this outcome since the Tick Size Pilot stocks in bin 1 and bin 2 still saw market quality improvements, implying that narrow tick concerns didn't yet dominate the effect on these stocks, it is unlikely that the smaller tick size reduction associated with this Rule would lead to small tick problems diminishing market quality.

Additional Sources of Information: Commenters also suggested additional settings to identify when stocks are trading with the optimal number of ticks intra-spread. One commenter suggested that the Commission should have considered the European Union's tick size approach,

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1286 One academic theoretical paper suggests that having a two-tick quoted spread is optimal. See Sida Li & Mao Ye, Discrete Prices, Discrete Quantities, and the Optimal Price of a Stock (working paper Mar. 8, 2021, revised Jul. 7, 2023), available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3763516 (retrieved from SSRN Elsevier database). The paper suggests that stocks reach their optimal price whenever the quoted spread is two ticks wide. While the paper advocates for a lower tick size, particularly for tick-constrained stocks, the two-tick quoted spread conclusion is the result of a highly stylized trading model which does not take into account pertinent factors from outside the model which likely affect quoted spreads such as considerations of time priority and pennying concerns. Conditional on there being non-infinitesimal tick and round-lot sizes, their model suggests that a two-tick wide quoted spread is optimal. Otherwise, their model suggests an optimal policy choice of infinitesimal tick and round-lot sizes.

associated with MiFID II, which assigns one of over 20 variable tick sizes based on trading price and number of transactions per day, and Japan's tick size approach.[1287]

One paper on the European experience cited by commenters studied the effects of a new tick regime introduced by MiFID II on 511 stocks listed on Euronext Paris and found results similar to the TSP analysis in Table 8.[1288] In this study the authors point out that MiFID II led to tick size increases for 339 stocks and decreases for 82 stocks. Tick increases were followed by a widening of the quoted spread, an increase in depth near the top of the book, and a reduction in message traffic. Like the TSP analysis in Table 8, tick decreases were followed by a narrowing of quoted spreads, reductions in depth at the top of the book, and an increase in message traffic; for a relatively wide price interval, depths remained unchanged. Table 8 finds a reduction in the cost of a round lot trade when low quoted spread securities experience a reduction in the tick; similarly, the analysis of MiFID II documents a reduction in transaction costs when the tick is reduced and the size of the trade is held constant. In sum, the effects of a change in tick size on Euronext Paris largely mirror the effects documented with the TSP in Table 8, indicating that the Commission's analysis is documenting a generalizable phenomenon.

However, there are several limitations to using studies of Europe's tick size regime. Research based on experience with the E.U. regime is difficult to apply to this Rule because the criteria in this Rule for determining the tick size is the TWAQS, which is not a factor in the European setting. When tick sizes change in Europe, it is due to changes in price or trading

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1287 See CCMR Letter at 27.

1288 Autoritè Des Marchès Financiers (AMF), MiFID II: Impact of the New Tick Size Regime (March 2018), available at https://www.amf-france.org/sites/institutionnel/files/contenu_simple/lettre_ou_cahier/risques_tendances/MiFID%20II%20Impact%20of%20the%20New%20Tick%20Size%20Regime.pdf. This paper was cited in Nasdaq Letter I at 10; Tradeweb Markets Letter at 4; Robinhood Letter at 50-51; and IEX Letter I at 13.

volume which can simultaneously affect quoted spreads. The Commission is unaware of research using the tick sizes associated with MiFID II to identify the thresholds, in terms of quoted spread, where a stock likely benefits or is harmed by a modification of the tick size, and the commenter did not provide such research.

In addition, it could be difficult to apply such analysis to the U.S. setting due to a number of structural differences between the European market and the U.S. markets. Key among these is that European financial markets are not as integrated as the U.S. national market system. For example, there is currently no consolidated tape or requirement to route orders to the exchanges with the best prices in the E.U.[1289] This fact can affect inference in the context of analyzing this Rule because it means that existing studies of the effect of the European tick size regime are generally limited to one exchange (e.g., the London Stock Exchange).[1290] This can be a significant limitation when trying to apply insights from European studies to U.S. markets because, as has been found in existing research, using data from one exchange as compared to across all exchanges can lead to opposite market quality effects being documented.[1291]

As discussed above, a commenter also suggested that the Commission should have considered the effects of tick size modifications in Japan to inform the appropriate thresholds for the tick size change considered in this Rule.[1292] The Tokyo Stock Exchange (TSE) assigns tick

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[1289] See Rule 611 and supra note 226.

[1290] See, e.g., Eros Favaretto et al., Impact of MiFID II Tick-Size Regime on Equity Markets—Evidence From the LSE, 29 EUR. FIN. MGMT. 109, 109-149 (2023).

[1291] For example, Chung, et al., supra note 1209, which uses data from all US exchanges, and Griffith and Roseman (2019), supra note 1002, who use data from just Nasdaq produce opposite findings in some areas concerning the effect of the TSP on various dimensions of market quality – i.e., the effect of the TSP on the cost to trade very large orders.

sizes ranging from 0.1 (JPY) to 100,000 (JPY) depending on the price of the security in question. Stocks in the TOPIX 100 index have a different tick size schedule than do other securities. Again, making inferences from the TSE to this Rule is difficult due to the structural differences between the way that the tick sizes operate. On the TSE, tick sizes are price determined, under this Rule, tick sizes depend on the prevailing quoted spread.[1293] Additionally, trading in Japan is largely consolidated on the TSE, whereas in the United States it is considerably more fragmented suggesting the same issue relating to inference as can occur with the European studies.

In considering the experience in the Japan markets, one study used the median quoted spread to divide stocks on the TSE and examines market quality.[1294] These researchers find that a tick size reduction reduces quoted spreads for both the narrower and wider quoted spread stocks – although the effects are considerably larger for narrower quoted spread stocks. The results of this study are consistent with those in this release as it documents that for some stocks, particularly those with narrower quoted spreads, reducing the tick size can reduce quoted spreads. This study also documents that depth at the best prices also tends to decrease with a smaller tick. However, this study has limitations in the context of using it to determine the optimal tick to quote size ratio because the authors do not separate stocks using a nominal quoted spread threshold which would allow inference about how stocks with quoted spreads above or below a certain threshold were affected by the tick size. Rather, they simply bifurcate the sample in half which doesn't apply a specific quoted spread threshold and so it is difficult to discern from their study what the optimal tick to quoted spread threshold will be.

[1293] See infra section VII.D.1.b.iii for additional discussion of price as a determinate of tick-constrained securities and thus of the tick size.

[1294] See Ingrid M. Werner et al., Tick Size, Trading Strategies and Market Quality, 69 MGMT. SCI. 3818 (2023).

One commenter presented two industry studies based on reverse stock splits showing large reductions in percentage spreads and hence trading costs following reverse splits.[1295] When a stock undergoes a reverse split, its share price rises. All else equal, one would expect quoted spreads to widen in proportion so that the trading cost remains constant as a percentage of price. If the stock is tick-constrained, however, a reverse split causes the current penny tick to become lower relative to the (higher) post-split price, relieving the tick constraint and reducing trading costs as a percentage of the trade amount.[1296] One study presented evidence from GE's eight-for-one reverse split, and showed that trading costs—as measured by the quoted spread as a fraction of the stock price—fell 75%.[1297] Another study presented evidence from five tick-constrained ETPs that underwent a five-for-one reverse split; the study found that trading costs for these ETPs declined substantially.[1298] These studies are consistent with results in Table 8 above—when the tick constraint is relaxed, the quoted spread becomes smaller relative to the share price, thereby reducing transaction costs.

Some commenters provided explicit specifications of what they stated is the optimal tick to quoted spread range.[1299] These commenters presented evidence and views typically suggesting

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1295 See Why GE's basis point spread was four times higher before its reverse split attached to MEMX Letter at 43-63; see also The Tick Size Debate Revisited attached to MEMX Letter at 64-70.

1296 See supra note 1064 for a numerical example showing the effect that a reverse split has on transaction costs for a tick-constrained stock.

1297 See Why GE's basis point spread was four times higher before its reverse split attached to MEMX Letter at 43-63. Chart A of the study indicates that the reverse split caused the average quoted spread as a fraction of the share price to decline from approximately 8 basis points to 2 basis points.

1298 See The Tick Size Debate Revisited attached to MEMX Letter at 64-70. The appendix of the study indicates that the reverse split caused transaction costs to fall by more than 40% for tick-constrained ETPs, with some ETPs experiencing cost reductions of 80%. The study further indicates at 66 that similar results, "can be achieved by amending the tick regime to simply allow more granular prices, without the need to change the price of the security in question."

1299 Nasdaq Letter I at 8 provides some empirical analysis suggesting that 2 to 3 or maybe 2 to 4 ticks intra-spread is optimal. Pragma Letter at 1 uses data from stock splits to suggest that 1.5 to 4 ticks is optimal and that 2 to 4 ticks intra-spread may be the optimal range. Given the analysis of these commenters, analysis presented here, and additional research,[1300] the Commission believes that the amendments, which only reduce the tick size when prevailing quoted spreads fall below $0.015, are likely to improve market quality for these stocks on average and are unlikely to lead to detrimental pennying or complexity concerns. Specifically, while there will likely be less depth at the NBBO and at each price level, quoted and effective spreads are likely to decline such that the cost of executing small and medium size trades will likely decline. Pennying is unlikely to predominate. This is because, at most, the smaller tick size will result in 3 ticks intra-spread. The analysis contained in this release, additional research,[1301] and commenters tend to agree that pennying is unlikely to be a dominant effect at 3 ticks intra-spread.[1302] In fact, 3-ticks intra-spread falls in the middle of the 2 to 4 ticks intra-spread suggested as potentially optimal by many commenters.[1303]

The narrower quoted spreads from a smaller tick size may result in fewer opportunities for price improvement by retail wholesalers, which may cause execution quality for wholesalers as measured by price improvement statistics to appear worse. However, the prices that retail

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that more than 4 is too many. RBC Letter at 3 cites research that 2 ticks intra-spread is optimal. CCMR Letter at 23 cites academic research suggesting 2 ticks intra-spread may be optimal. XTX Markets Letter at 4 suggests 2 to 4 ticks intra-spread as optimal. MMI Letter at 5 suggests 3-9 as optimal. IEX Letter I at 14 states for 2 to 4 ticks intra-spread. Budish Letter at 5 indicating that two or fewer is too few and suggesting that 2 to 4 may be reasonable. Harris Letter at 7 suggesting 2 ticks intra-spread.

1300 See Barardehi et al., supra note 231.

1301 Id.

1302 One commenter stated that at 4 or more ticks intra-spread hidden orders become more prevalent which can harm market quality. See IEX Letter I at 12. However, the smaller tick size implemented by the Rule will not result in 4 or more ticks for the associated stocks and so, even if true, the effect suggested by the commenter is unlikely to play a large role in market quality because the rule will not result in 4 or more ticks intra-spread for stocks receiving the $0.005 tick size.

1303 Other commenters expressing support for a $0.005 tick for stocks with narrow quoted spreads include Schwab Letter II at 6, STA Letter at 7, XTX Letter at 4.